In Google Maps, adding traffic data creates an informative digital twin at a global scale. This traffic data, gathered from fixed sensors and mobile devices, is rendered by color-coding streets according to traffic density and speed as shown in figure 1.6. This pattern of overlaying data layers on map views is one of the most successful design patterns in informative digital twin development.

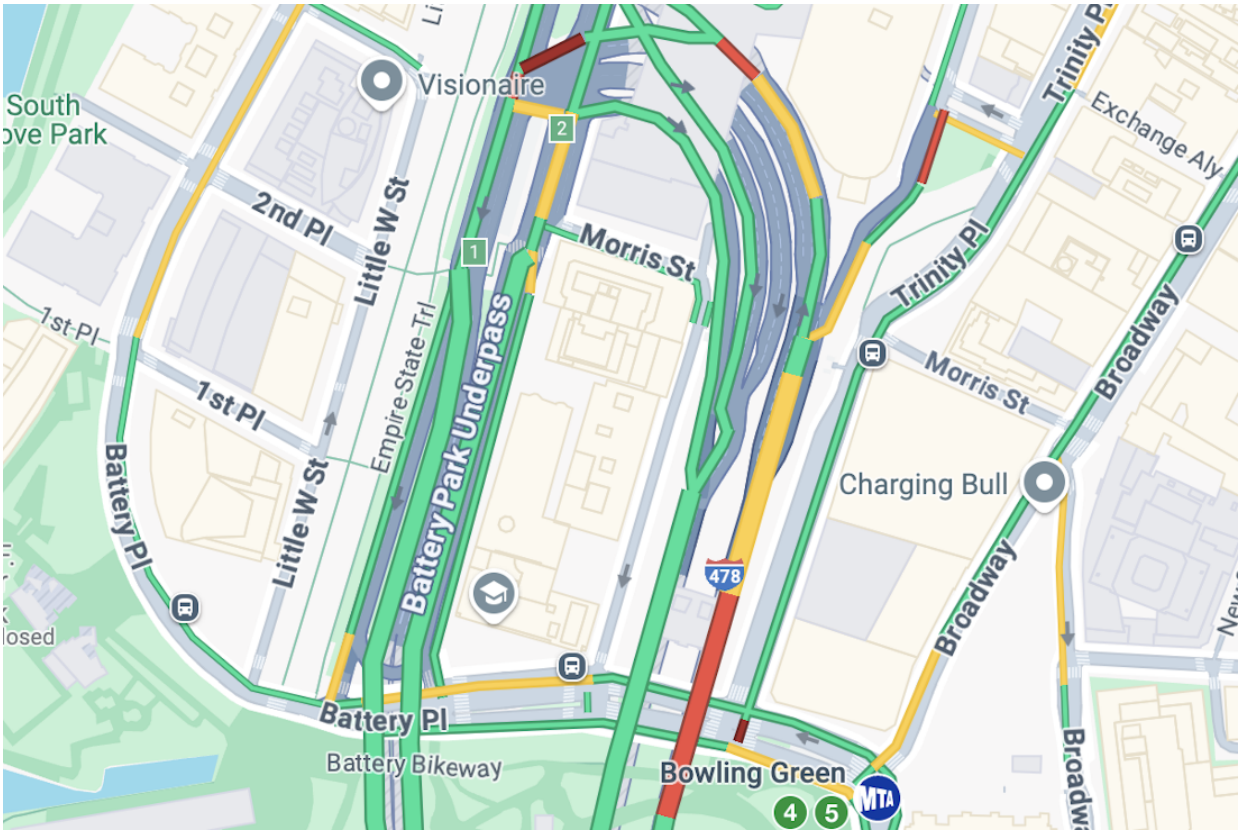


Figure 1.6 Google Maps view of lower Manhattan with realtime traffic congestion data overlaid is an example of an informative digital twin. Imagery © 2025 Google, Map data © 2025, Map data © 2025 Google.

REAL WORLD EXAMPLE: TEMPERATURE MONITORING

Other informative digital twins offer laid-out dashboards with graphs, gauges, dials, and other components. Figure 1.7 shows one displaying temperature fluctuations over several days inside and outside a building and how the indoor temperature is correlated to that outdoors. Behind this simple visualization lies a network of IoT sensors streaming data into the cloud that is then contextualized (for example, linking an individual sensor to a specific